NAND to Tetris-I

Lab 2: Sequential Logic & Memory Hierarchy

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Q1. Q-2 (5 marks)

CHIP Register { IN in[16] , load; Out out[16]; Parts: Bit(in = in[0], load=load, out=out[0]); Bit(in = in[1], load=load, out=out[1]); Bit(in = in[2], load=load, out=out[2]); Bit(in = in[3], load=load, out=out[3]); Bit(in = in[4], load=load, out = out[4]); Bit(in = in[5], load=load, out = out[5]); Bit(in = in[6], load=load, out=out[6]); Bit(in = in[7], load = load, out=out[7]); Bit(in = in[8], load = load, out = out[8]); Bit(in = in[9], load = load, out = out[9]); Bit(in = in[10], load = load, out = out[10]); Bit(in = in[11], load = load, out = out[11]); Bit(in = in[12], load = load, out = out[12]); Bit(in = in[13], load = load, out = out[13]); Bit(in = in[14], load = load, out = out[14]); Bit(in = in[15], load = load, out = out[15]); }

Q2. Q-4 (5 marks)

CHIP RAM8 { In in[16], load, address[3]; Out out[16]; Parts: DMux8way(in=load, sel=address, a=10, b=11 ,c=12 ,d=13, e=14, f=15 ,g=16, h=17, Register(in=in, load=10, out=r0); Register(in=in,load=11, out=r1); Register(in=in,load=12, out=r2); Register(in=in,load=13, out=r3); Register(in=in,load=14, out=r4); Register(in=in,load=15, out=r5); Register(in=in,load=16, out=r6); Register(in=in,load=17, out=r7); Mux8way16(a=r0, b=r1, c=r2, d=r3, e=r4, f=r5, g=r6, h=r7, sel=address, out=out); } CHIP RAM64 { In in 16, load ,address[6]; out out[16]; PARTS: DMux8way(in=load, sel=address[3..5], a=10 ,b=11, c=12, d=13, e=14 ,f=15, g=16, h=17); RAM8(in=in, load=10, address=address[0..2], out=r0); RAM8(in=in, load=11, address=address[0..2],out=r1); RAM8(in=in, load=12, address=address[0..2],out=r2); RAM8(in=in, load=13, address=address[0..2],out=r3); RAM8(in=in, load=14, address=address[0..2],out=r4); RAM8(in=in, load=15, address=address[0..2],out=r4); RAM8(in=in, load=16, address=address[0..2],out=r5); RAM8(In=in,load=17, address=address[0..2],out=r6); Mux8way16(a=r0, b=r1, c=r2, d=r3, e=r4, f=r5, g=r6, h=r7, sel=address[3..5], out=out); }

Q3. Q-6 (4 marks)

CHIP PC { In in[16], load, inc, reset; Out out[16]; PARTS: Inc16(in = out, out=incout); Mux16(a=out , b=incout , sel=inc , out=incSelected); Mux16(a=incSelected, b=in, sel=load, out=loadSelected); Mux16(a=loadSelected, b=false, sel=reset, out=next); Register(in=next, load=true, out=out); }

Q4. Q-7 (10 marks)

Files: Q7_Lab2_Sequential_Logic_Memory_Hierarchy_Aditya_Agrawal.pdf

Q5. Q-1 (5 marks)

CHIP Bit { IN in, load; Out out; Parts: Mux(a=out, b=in , sel=load, out=next); DFF(in=next , out=out); }